

The selected construction, at three levels

[Info Docs] Architecture Reference and Workflow Diagrams for PR #277

THE NUMERICAL CONTRACT

513 retained PR259 cohorts + seven delayed-inverse intervals + the same 25 PR258/260 retimings applied once.
Conditional kappa: 0.000711045547624400.
Literal stock 867,045; 19,786,600 paid children.

HOW TO READ THE CONNECTIONS

21 current blocks and 53 typed block dependencies. All 87 original hierarchy IDs are reconciled separately.
Architecture views show high-level process and compiler flow. Inherited all-size internals remain contracts.

Full integer-product hierarchy and all 87 original blocks

Functional: source-bound local supplier

Functional: global assembly and exact closure

Block catalog: what, why, ports, requirements and proof

Requirements and source traceability

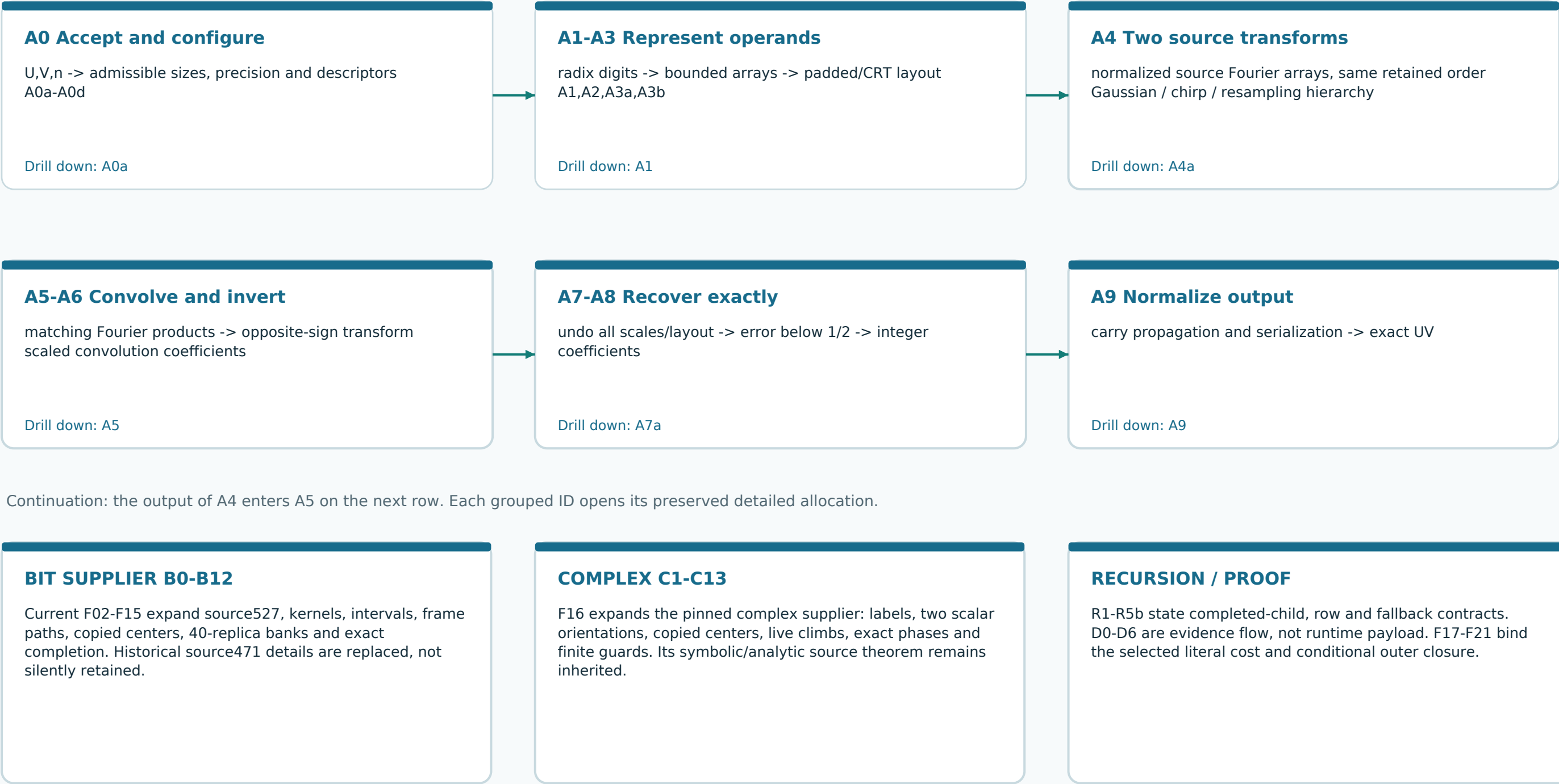
Exact source anchors and hashes

Coverage, heritage and reproduction boundary

Navigation: clickable blocks open the catalog; source IDs open exact proof anchors; every page returns here.

Functional hierarchy | integers to exact product

The selected hybrid replaces the BIT supplier inside this retained conditional integer-multiplication architecture.



Functional hierarchy | source-transform internals

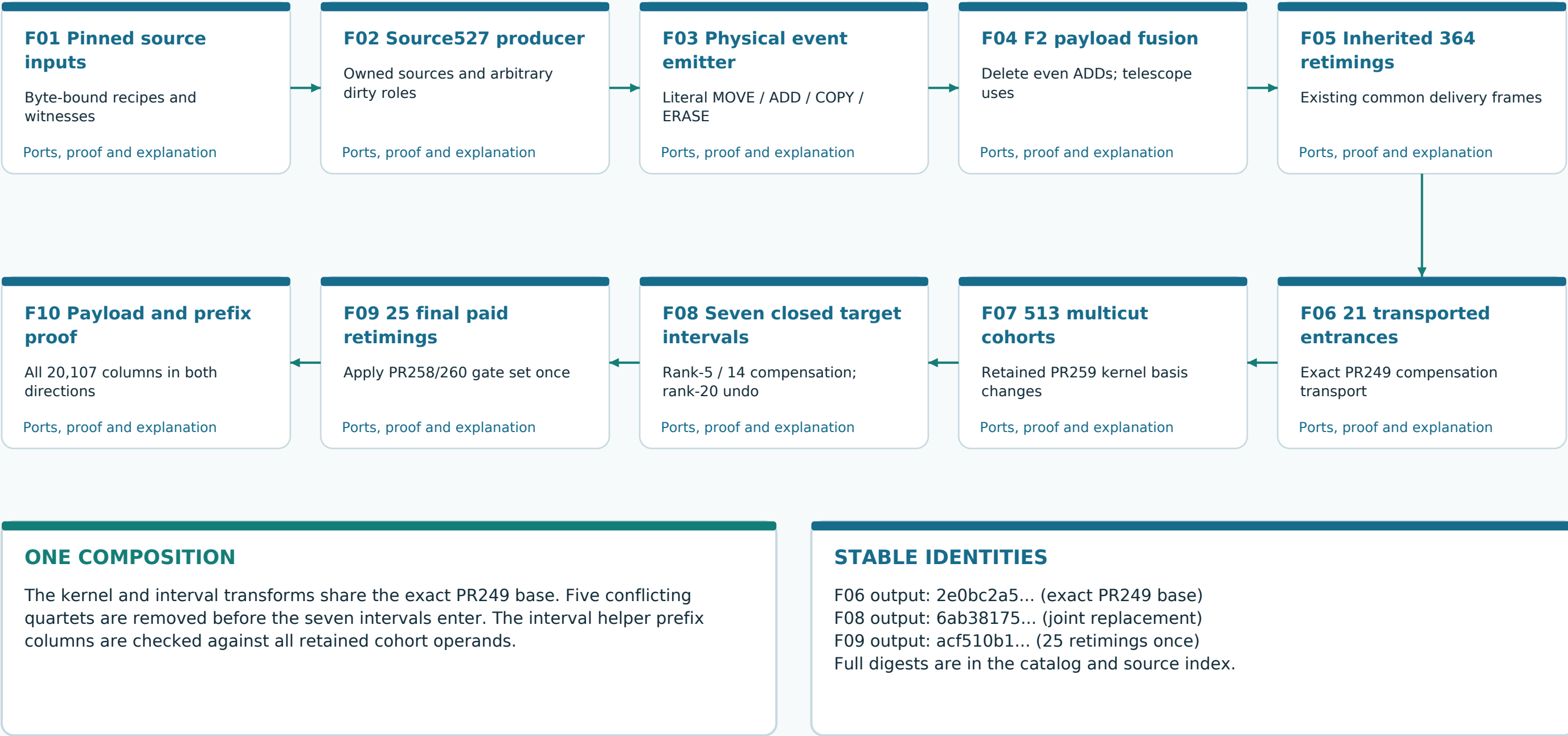


Every original A/B/C/R/D ID appears below with its typed ports, why it exists, current allocation and exact scope.

Unexpanded all-size numerical routines and theorem primitives remain explicit contracts. No transistor/gate-level or all-size implementation completeness is claimed.

Functional architecture | local supplier

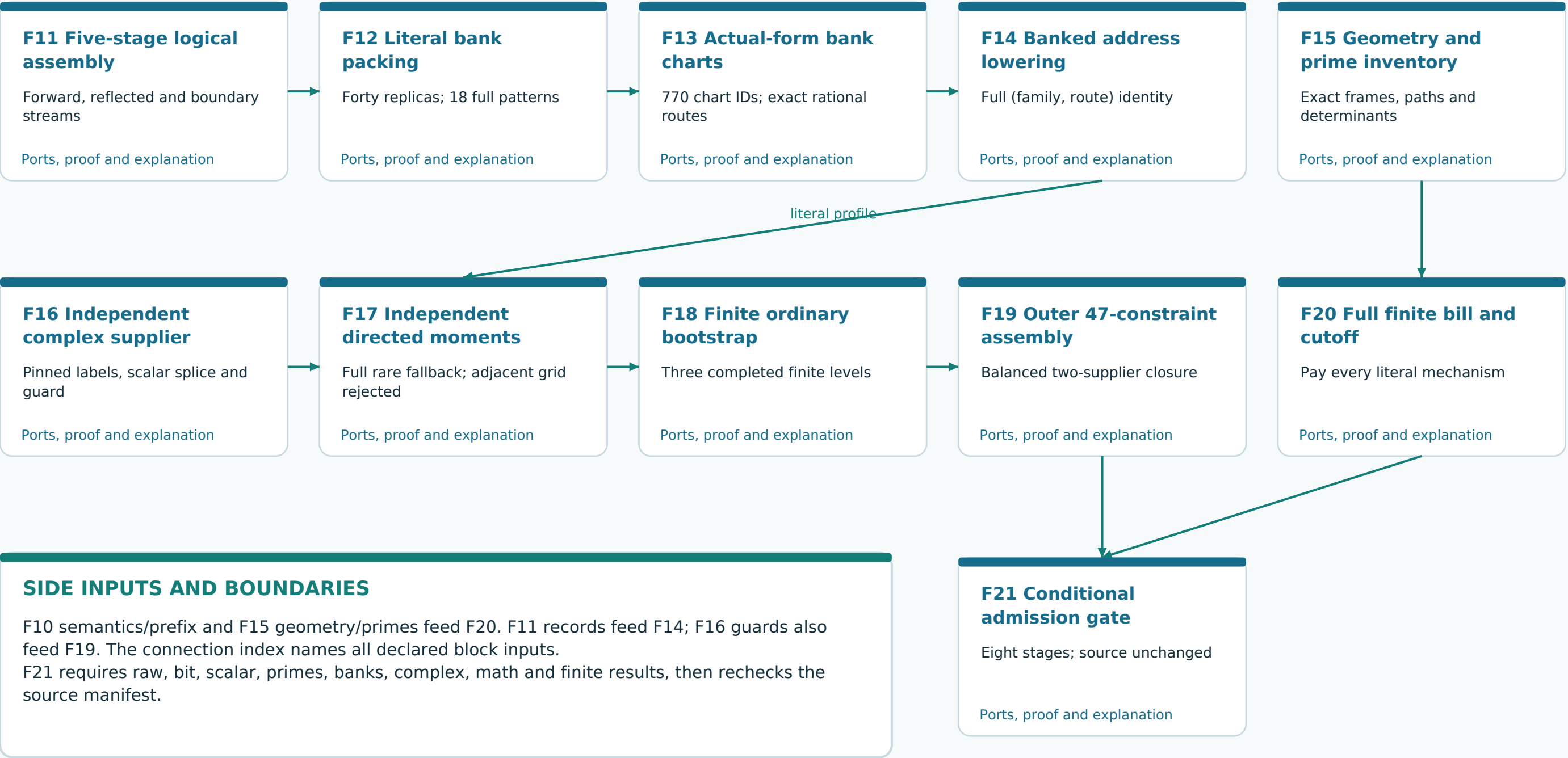
Pinned source recipes -> literal records -> selected compatible transformations -> independently checked payload.



Cross-edges: F01 supplies all selections; F06 supplies the interval witness base. F09 also feeds F11/F15 on the next sheet.

Functional architecture | complete conditional closure

The bit and complex suppliers remain distinct. Scalar, geometric, bank, moment and finite proof ports must all close.



The arrows are compiler/proof dataflow. No local optimization or individual PR contributes an independent additive kappa gain.

Functional blocks | F01 + F02

F01 Pinned source inputs

WHAT

Read the selected package manifest, source527 recipes, PR259 candidate gzip, exact interval and retiming witnesses, and the independent complex certificate. Reject missing, extra, changed or symlinked package files.

WHY

Every later identity and price must refer to one reproducible construction, rather than a compatible-looking saved histogram.

INPUT PORTS

package: immutable files and MANIFEST.json

OUTPUT PORTS

pins: manifest digest and complete file census; recipes: exact witness bytes

REQUIREMENTS

R01 R11

INVARIANTS

Manifest SHA256 is 9f6d8637b9a119cfe4566a02ab2d704f23c502e5e4357b40c28ef85a3fd4de89. Assertions enabled; no saved execution receipt is a verifier input.

HERITAGE

Current composition; imported sources retain their own notices.

PROOF / IMPLEMENTATION

S01 verify.py : 26

S02 PR259-PROVENANCE.md : 1

F02 Source527 producer

WHAT

Reconstruct the fixed source program, physical ownership, source aliases, target aggregation, rank/echelon transforms, borrowed roles and exact frame registry. The 527 source aliases are inherited, not newly searched by this documentation build.

WHY

The selected transformations need a fully defined semantic producer and ownership map. Arbitrary dirty contents cannot be replaced with zero-filled conveniences.

INPUT PORTS

recipes: pinned source and selection files

OUTPUT PORTS

context: W, C, regs, borrow, SOURCE_TEXT; sourceword: defining scalar program

REQUIREMENTS

R01 R02 R03

INVARIANTS

v=1,760; h=24; 16,587 independent helper roles; 20,107 formal local columns. Producer context and defining signed decoder remain unchanged by the new hybrid.

HERITAGE

Source527 integrates PR210, PR230, PR216, PR207, PR211, PR200 and retained dependencies; exact source for this composition is PR249.

PROOF / IMPLEMENTATION

S03 prepare.py : 12

S04 source527/word.py : 1

Functional blocks | F03 + F04

F03 Physical event emitter

WHAT

Instrument the reconstructed source program to emit six-integer local records, exact required-use frame paths and tagged scalar events. A single temporary serves 24 copied-center episodes; each has a rank-22 paid copy and 220 scatter reads.

WHY

A literal physical word binds algebra to movement cost and copied-center lifetime. A scalar identity alone does not price the construction.

INPUT PORTS

context: owned roles, frames and source text

OUTPUT PORTS

word: six-field physical records; paths: initial/final frames, required uses, categories

REQUIREMENTS

R02 R03 R04

INVARIANTS

COPY source is immutable during its window; no stale temporary read or inverse erasure. Every positive MOVE increment and center copy contributes to the paid histogram.

HERITAGE

PR234 five-stage/source-bound physical interface, adapted to source527.

PROOF / IMPLEMENTATION

S05 code/physical527.py : 11

F04 F2 payload fusion

WHAT

Remove even-coefficient payload additions, preserve all surviving odd additions in order, and fuse monotone frame moves only between surviving uses and required boundaries. Rebuild forward and reflected paid paths.

WHY

Payload scalars are binary even though address geometry is rational and uses odd primes. Removing an F2 identity can eliminate otherwise unnecessary alignment work.

INPUT PORTS

word: defining physical records

OUTPUT PORTS

word: parity-filtered physical records; ledger: recomputed forward/reflected MOVE widths

REQUIREMENTS

R02 R03 R04 R06

INVARIANTS

1,783,196 even ADDs removed; 786,621 surviving ADDs before later transforms. COPY/ERASE, old endpoints, stage boundaries and odd-prime address arithmetic remain intact.

HERITAGE

PR244 parity-filtered/common-frame source lineage; primary bit scalar contract retained.

PROOF / IMPLEMENTATION

S06 parity_transform.py : 62

S07 proof/PARITY-CONTRACT.md : 20

Functional blocks | F05 + F06

F05 Inherited 364 retimings

WHAT

Apply the already selected 364 common-frame event retimings and reconstruct operand paths and ledgers. These are part of the exact PR249 base and are distinct from the final 25 scalar-keyed retimings.

WHY

Use legal later common frames to combine paid movement while preserving scalar order, data endpoints and copy lifetimes.

INPUT PORTS

word: parity-filtered records; selection: retiming-selection.json

OUTPUT PORTS

word: inherited-retimed records; ledger: exact local histogram

REQUIREMENTS

R03 R04

INVARIANTS

364 events, including 353 unborrowed pair mixes, use their existing common delivery frames. No change to input/output frames or copied-center lifetimes.

HERITAGE

PR244 and exact PR249 base lineage.

PROOF / IMPLEMENTATION

S08 retiming_transform.py : 65

S09 raw_ledger.py : 190

F06 21 transported entrances

WHAT

Apply the 21 selected PR249 gauge entrances, checking exact compensation transport, untouched helpers before the semantic cut and target-only prefix noise. Keep producer and execution contexts separate.

WHY

Raise selected helper entrances only when their earlier payload effect can be transported exactly, reducing residual movement without changing the source producer.

INPUT PORTS

word: inherited-retimed records; selection: regauge-selection.json

OUTPUT PORTS

base: exact PR249 physical word; entrances: actual nonzero input frames

REQUIREMENTS

R02 R03 R04

INVARIANTS

21 added entrances, total added entrance rank 325. Base raw SHA256 2e0bc2a5c54cf8caf26fa30a1a2949702d842f620b28023ca2ba59140ecbed1a. The sixteen extra PR251 entrances are absent.

HERITAGE

PR249 96495746c786d6d0339dbb38c7f553d4af3f88ed, eumemic.

PROOF / IMPLEMENTATION

S10 regauge_transform.py : 128

S11 raw_ledger.py : 216

Functional blocks | F07 + F08

F07 513 multicut cohorts

WHAT
Read all 518 pinned PR259 candidates, remove the five conflicting rank-11 quartets and install the remaining 513 exact entrance bases. At each declared cut emit donor += pivot, and emit literal inverses at the common full endpoint.

WHY
Share a pivot contribution through an admissible kernel basis, replacing many target reads with paid, source-bound basis changes.

INPUT PORTS
base: PR249 word; candidates: 518 exact PR259 cohorts; selection: five removed pivots

OUTPUT PORTS
cohorts: 513 retained groups; word: kernel basis and inverse events; entrances: cohort pivot frames

REQUIREMENTS
R01 R02 R03 R04

INVARIANTS
Removed pivots: 17207, 17222, 17246, 17255, 17273. Unique pivot support; no pivot/donor collision; basis contained in each actual first-use frame. Within each cut use (dimension,pivot) order; the newly installed cohort-plus-interval entrance rank is 1,422.

HERITAGE
PR259 a53ec0ca4d59634f16630dd16adf035c9f12595c, Dugongue; candidate gzip included byte-for-byte.

PROOF / IMPLEMENTATION
S12 hybrid_transform.py : 19
S13 hybrid-selection.json : 4
S14 PR259-PROVENANCE.md : 3

F08 Seven closed target intervals

WHAT
For each interval precondition three targets against a pivot at frame zero, compensate the pivot from low/high helpers at nested rank-5 and rank-14 frames, and undo the target preconditioner at a legal common rank-20 frame.

WHY
The delayed inverse changes the paid movement partition. Restricting the inverse to the high frame would describe a different family and miss this selected construction.

INPUT PORTS
base/cohorts: surviving event word; intervals: 14 helpers, 28 target IDs and exact bases

OUTPUT PORTS
word: seven eight-ADD local replacement blocks; entrances: seven rank-5 and seven rank-14 frames

REQUIREMENTS
R02 R03 R04 R06

INVARIANTS
Every interval has an exact six-column integer identity preserving arbitrary old targets. The 14 signed prefix columns vanish on every retained-cohort operand at all four cuts. No external helper/source observer survives inside a closed interval; newly installed cohort-plus-interval entrance rank is 1,422.

HERITAGE
New source-bound hybrid composition of delayed-inverse intervals with PR259; inherited target-recoding and bank ideas retain credit.

PROOF / IMPLEMENTATION
S15 hybrid_transform.py : 92
S16 PROOF.md : 16

Functional blocks | F09 + F10

F09 25 final paid retimings

WHAT

Match 25 unique scalar keys against their exact old basis hashes, move those events to nominated existing successor frames, and re-emit all operand paths. This adapter runs after the combined kernel/interval replacement.

WHY

The retiming is a compatible physical improvement, not a second copy of either contributor's same-gate result.

INPUT PORTS

word: replacement raw 6ab38175...; selection: retiming25-selection.json

OUTPUT PORTS

finalword: raw acf510b1...; ledger: 96,117 calls; rank mass 432,755; paths: required-use frame paths after every selected transform; registry: actual final frame bases and annihilators; entrances: exact nonzero helper input frames; context: final execution W/C/role registry; raw: fully rebound selected source ledger

REQUIREMENTS

R01 R03 R04

INVARIANTS

No new chart; scalar/COPY projection and all endpoints byte-identical. Histogram delta: 2:-2, 4:+24, 5:-48, 6:+25, 11:-1, 13:+2, 17:-1. PR258 and PR260 identify the same 25 gates; apply once.

HERITAGE

PR258 cf82ca09322d1327ab887cfd1cce97591b8d3cfa (amanforindia); PR260 87aa555a455cca84a3e3296ced2440623d39fc4d (Rohan Arun).

PROOF / IMPLEMENTATION

S17 retiming25.py : 22

S18 hybrid_transform.py : 146

F10 Payload and prefix proof

WHAT

Independently replay the emitted word and literal reverse on all formal F2 source, old-target and arbitrary-dirty columns. Separately compute cancellation-free signed-lift prefix majorants. The unchanged defining source receives its own signed decoder tests and 25 hostile controls.

WHY

F2 equality, signed source evidence and chosen-lift growth are different claims. Keeping them separate prevents an invalid old-integer-endpoint requirement or an uncharged precision assumption.

INPUT PORTS

finalword: physical records; sourceword: unchanged defining program

OUTPUT PORTS

semantics: all-column F2 identity and inverse; prefix: 73,095 forward; 3,396,290 inverse

REQUIREMENTS

R02 R06 R11

INVARIANTS

760,383 weighted events; 763,023 unit-expanded additions locally. All source and dirty columns restored; arbitrary old target contents preserved. No assertion that the new chosen integer lift equals the old signed decoder endpoint.

HERITAGE

Inherited source decoder plus current hybrid all-column F2 and prefix proof.

PROOF / IMPLEMENTATION

S19 scalar_check.py : 30

S20 scalar_check.py : 54

Functional blocks | F11 + F12

F11 Five-stage logical assembly

WHAT

Lower the same local word into five stage-specific renamed streams with orientations F,R,F,R,F. After stages 1,2,4, emit exact idle paths and six bridge families; after all stages emit tagged entrance completions and final complete-stream exchanges.

WHY

Compose the local supplier into the actual width-120 global architecture, binding matrix routes and role names rather than merely multiplying a local histogram by five.

INPUT PORTS

finalword: local physical records; paths: frame projectors and entrances

OUTPUT PORTS

global: eight-field logical records and phase schedule; completions: 2,827 exact entrance tags; ledger: unbanked global histogram

REQUIREMENTS

R02 R03 R04 R05

INVARIANTS

Active pairs: (X1,Y1),(Y1,X1),(X1,Y1),(Y2,X2),(X2,Y2). Every reverse stage copies freshly from the original center; 120 center episodes per replica. Cover expansion is phase-major; terminal exchange is a paid complete-stream permutation.

HERITAGE

Henry Grant / hcg890 PR234 five-stage architecture and finite interfaces.

PROOF / IMPLEMENTATION

S21 code/global_lowering.py : 66

S22 portable_bit.py : 15

F12 Literal bank packing

WHAT

Partition residual helper widths into 18 exact width-120 bank patterns across forty literal replicas. Enumerate every original role occurrence, sorted by residual family, and assign it to one contiguous block in a stage-private bank.

WHY

Complete banks can replace independent completion calls only when all coordinates and all dirty-role ownership are covered with integer capacity.

INPUT PORTS

entrances: actual helper role census and gauges; global: stage/replica coordinates

OUTPUT PORTS

packing: 117,089 banks per stage; stock: 585,445 helper-bank and 281,600 data families

REQUIREMENTS

R04 R05

INVARIANTS

867,045 live families; external work separate. 3,317,400 stage/replica/role assignments; no rounded, unused or fractional bank capacity. All 18 patterns partition 120 coordinates exactly.

HERITAGE

Completed-bank lineage includes PR197 and PR237 width-120 precedent; this exact 40-replica packing/assignment is current integration.

PROOF / IMPLEMENTATION

S23 bank_template.py : 11

S24 bank_check.py : 49

S25 BANK-PROOF.md : 9

Functional blocks | F13 + F14

F13 Actual-form bank charts

WHAT

For every consumed entrance frame construct residual columns from 15a - sum(a)1, append the actual source basis, compute both exact chart inverses and replay rational factors. Build $N=(block+1) \Pi \text{ embed_stage}(B^{-1})$.

WHY

A correct chart must use the actual form 9I-J. Generic construction handles entrances for which a norm-four rank-one shortcut is false.

INPUT PORTS

registry: actual source bases and annihilators; packing: assigned stage/window/block

OUTPUT PORTS

charts: 770 frame-ID charts, 459 distinct entrance bases; normalizers: N and exact inverse factor program

REQUIREMENTS

R03 R04 R05 R07

INVARIANTS

$BA=AB=dl$; each projector acts correctly on every chart column. At most 576 chart factors; conservative normalizer tariff remains 838. Outside-column witnesses distinguish full routes even when chart bytes coincide.

HERITAGE

Actual-form chart implementation in current bank integration; inherited common-ancestor chart theorem remains conditional.

PROOF / IMPLEMENTATION

S26 bank_check.py : 97

S27 bank_template.py : 68

S28 BANK-PROOF.md : 39

F14 Banked address lowering

WHAT

Rewrite only stream operands in the logical records to full banked addresses. Preserve every scalar, frame, projection, stage and reversal field. Match all tagged completions, prove the complete bank endpoint, then subtract their exact counts.

WHY

Family equality is insufficient for alias testing. Completion savings require a complete bank proof and cannot be implemented as silent opcode suppression.

INPUT PORTS

global: logical records and completion tags; packing/charts: exact addresses and normalizers

OUTPUT PORTS

bankword: actual callable banked streams; literal: 19,786,600 children; mass 103,869,400

REQUIREMENTS

R02 R04 R05

INVARIANTS

Helper class maps $d*\tau_i$ to $d*\tau_i*N^{-1}$, in that order. All 200 complete stage/replica namespaces injective; ADD/COPY full addresses nonaliasing. Subtract exactly 2,827 tags per replica; retain ordinary children sharing widths 5,10,15,20,50.

HERITAGE

Current stage-private adapter; preserves PR234 logical semantics and inherited completed-bank obligations.

PROOF / IMPLEMENTATION

S29 bank_template.py : 130

S30 bank_template.py : 146

S31 bank_check.py : 65

Functional blocks | F15 + F16

F15 Geometry and prime inventory

WHAT

Check exact rational projectors, all relevant frame inclusions, reflected orientations, routes and terminal boundary identities. Rebuild the inventory of every actual physical or defining-source basis and evaluate exact Gram determinants and admissible-prime exclusions.

WHY

Equal dimensions or frame IDs do not prove equality of subspaces or legality over an eligible odd-prime address ring.

INPUT PORTS

context: actual bases, frames and routes; final/global: consumed inventory

OUTPUT PORTS

geometry: exact operator/path checks; primes: source-bound determinant inventory and six controls

REQUIREMENTS

R03 R07 R11

INVARIANTS

770 actual entrance frame IDs represent 459 distinct entrance bases; these counts must not be conflated. $q > 2^{80}$ after fixed exclusions; prime supply is an inherited theorem dependency. Every physical basis plus all defining-source and added-transform bases is covered.

HERITAGE

PR234 geometric interface plus source527/current entrance inventory; retained exact-prime framework.

PROOF / IMPLEMENTATION

S32 code/geometry527.py : 11

S33 prime_check.py : 20

F16 Independent complex supplier

WHAT

Reconstruct the pinned complex gcert/1 certificate, both scalar orientations, temporary lifetimes, label/scalar splice, all final climbs and finite precision/row guards. Feed its independent profile into the outer assembly.

WHY

The integer-multiplication closure uses both a bit supplier and a complex supplier. The complex theorem is not implied by a successful bit-word check.

INPUT PORTS

complexpins: certificate, source theorems and exact matrices

OUTPUT PORTS

complex: m=110, W=14,692, 361,405 children; guard: row coefficient 20,161; exact denominator grid

REQUIREMENTS

R04 R06 R10 R11

INVARIANTS

Rank mass 1,613,040; deficit 3,080; largest child 46. Three forward plus two backward calls use 7,577,864 real-component primitive steps. Denominators 1,2,3,6; no return rounding; source symbolic/analytic theorems remain dependencies.

HERITAGE

Jacob Sussman source f010392c923279e3dd59ef3aa5fedad23affc5fd; Avi Eisenberg / ikeboy PR193 helper and retained Claude assistance disclosures.

PROOF / IMPLEMENTATION

S34 portable_complex.py : 77

S35 portable_complex.py : 25

Functional blocks | F17 + F18

F17 Independent directed moments

WHAT

Price the literal banked bit histogram and unchanged complex histogram with outward exact-rational bounds. Independently cross-check the bit moment with the base-two engine and include the entire rare fallback allowance.

WHY

Concavity-based savings are only admissible with a rigorous strict gap and every paid residual, including the low-probability expensive branch.

INPUT PORTS

literal: bit histogram, $m=120$, $W=867,045$; complex: independent supplier profile

OUTPUT PORTS

rates: strict bit/complex moment enclosures; grid: next bit-grid exclusion

REQUIREMENTS

R04 R08 R09

INVARIANTS

Bit max child $50 < 120/2$; literal deficit 176,000. Fallback adds 10^{-16} times $32 \cdot m^2$ rank-one calls per paid child. Two independent interval engines agree; decimals are display/proposal only.

HERITAGE

Inherited weighted moment interface; independently implemented current literal repricing.

PROOF / IMPLEMENTATION

S36 math_check.py : 23

S37 moment.py : 55

S38 base_two_moment.py : 52

F18 Finite ordinary bootstrap

WHAT

Start with completed ordinary $a_0=384599/10^{10}$ and repeat $a_{(j+1)}=(1-c)c+c \cdot a_j$ exactly three times. Require every atom, borrowing, remainder and row-stock gap to remain positive.

WHY

The bit moment rate is converted to an actual completed ordinary supplier through a finite induction, rather than using an unattained limiting rate.

INPUT PORTS

rates: exact coarse bit c and inherited ordinary a_0

OUTPUT PORTS

ordinary: three-step exact chain and positive gaps

REQUIREMENTS

R08 R09

INVARIANTS

Every $a_j < a_{(j+1)} < c < 1-a_{(j+1)}$. Each stage includes an inherited-constant-dependent finite cutoff and old-leaf fallback.

HERITAGE

Retained ordinary-bootstrap and restored-row framework, freshly evaluated for this literal construction.

PROOF / IMPLEMENTATION

S39 math_check.py : 46

S40 finite_check.py : 130

Functional blocks | F19 + F20

F19 Outer 47-constraint assembly

WHAT

Combine the finite ordinary bit saving with the complex saving, row/recovery bridge, $\eta=10^{-12}$ and stopping fraction $\beta=10^{-9}$. Check all 47 strict inequalities and seven margins before choosing the admissible rational κ grid point.

WHY

A local helper saving is not the integer-multiplication exponent. The full coupled outer system determines which supplier and margin is binding.

INPUT PORTS

ordinary: completed bit saving; complex: rate and finite row/precision bridge

OUTPUT PORTS

bound: $\kappa=1777613869061/2500000000000000$; margins: 47 strict constraints and seven margins

REQUIREMENTS

R09 R10

INVARIANTS

κ decimal 0.000711045547624400; bit is binding. Adjacent κ grid point is rejected separately. Full all-size, recovery, symbolic and analytic interfaces remain conditional.

HERITAGE

Inherited balanced outer assembly, re-evaluated for current construction.

PROOF / IMPLEMENTATION

S41 outer.py : 16

S42 math_check.py : 53

F20 Full finite bill and cutoff

WHAT

Bind the selected word, tags, global streams, banks, geometry, prime inventory and moment profile. Add scalar primitives, movement, bridges, routes, chart normalizers, selectors, matrix preparation, copy/erase, fallback and restored-row costs.

WHY

Neither a better histogram nor a displayed finite coefficient can replace the complete accounting and full inherited-constant cutoff rule.

INPUT PORTS

final/global/literal: bound physical evidence; semantics/primes/rates: proof outputs

OUTPUT PORTS

finite: coefficient 62,644,197,182,478,801; prefix: 288305228198350917154735200000; cutoff: all inherited constants retained

REQUIREMENTS

R04 R06 R07 R08 R09

INVARIANTS

Coefficient below 2^{80} ; prefix is 98 bits and below 2^{104} . Selector allowance $667,542,305,600 < 2^{40}$; normalizer tariff 838. One-level max-child half-shrink holds; no recursive child hidden in a fixed constant.

HERITAGE

Current finite re-bill over inherited bit/row/copy primitives; exact theorem bytes remain pinned.

PROOF / IMPLEMENTATION

S43 finite_check.py : 59

S44 finite_check.py : 17

Functional blocks | F21

F21 Conditional admission gate

WHAT

Run all eight mandatory stages from inputs, check complete result coverage and expected rational kappa, then recheck the source manifest before emitting the certificate. Omitted stages must fail hostile controls.

WHY

The certificate must close every proof port and bind one immutable package. Documentation regeneration is not a substitute for scientific replay.

INPUT PORTS

raw: freshly reconstructed source ledger; bit: physical/global/geometry proof; scalar: F2, signed source and chosen-lift prefix proof; primes: current basis/determinant evidence; banks: callable full-address banked word and census; complex: pinned source, splice and finite guard; math: moments, bootstrap and outer47; finite: complete bill and cutoff

OUTPUT PORTS

certificate: conditional finite-construction result; verification: timings, stage coverage and immutable-source status

REQUIREMENTS

R01 R11 R12

INVARIANTS

Standalone verify.py executes all eight stages; output must be new and outside source. Recorded cloud admission freshly ran changed stages and reused byte/AST-bound unchanged signed-source and complex evidence. Mac standalone reproduction passes all eight stages with 143 source files unchanged; certificate SHA256 7eb58ec449c932b7092481746390c776365f302a742836cddef5643c0351802b. The construction remains mathematically conditional and is below the later PR266 frontier.

HERITAGE

Current source-bound verification orchestrator and separately scoped review/admission receipts.

PROOF / IMPLEMENTATION

S45 verify.py : 50

S46 verify.py : 37

Full hierarchy | blocks 1-6 of 87

A0a Read operands and lengths

WHY: Determine the exact input domain and operand lengths.

PORTS: in: Bits; out: Descriptors

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract
SYS-3	Exact historical proof/source		

A0c Generate descriptors

WHY: Make address, scale and phase choices explicit and reproducible.

PORTS: in: Descriptors; out: Descriptors

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract
SYS-3	Exact historical proof/source		

A1 Extract radix digits

WHY: Expose a convolution whose integer coefficients recover the product.

PORTS: in: Descriptors; out: Digits

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract
SYS-1	Exact historical proof/source		

A0b Select admissible sizes and precision

WHY: Choose sizes and precision that satisfy every later interface.

PORTS: in: Descriptors; out: Descriptors

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract
SYS-3	SYS-4	IF-5	Exact historical proof/source

A0d Select ordinary cutoff and guards

WHY: Route small cases to the attained ordinary leaf and retain finite guards.

PORTS: in: Descriptors; out: Descriptors

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract
SYS-3	Exact historical proof/source		

A2 Scale bounded inputs

WHY: Keep transform inputs within the declared numerical envelope.

PORTS: in: Digits; out: ScaledInputs

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract
SYS-1	Exact historical proof/source		

Full hierarchy | blocks 7-12 of 87

A3a Zero-pad operands

WHY: Prevent cyclic wraparound from corrupting ordinary convolution.

PORTS: in: ScaledInputs; out: PaddedInputs

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract
SYS-1	Exact historical proof/source		

A4a Gaussian expansion

WHY: Move source samples into the working transform grid.

PORTS: in: CRTInputs; out: WorkingTensor

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract	
SYS-1	SYS-4	IF-1	IF-5	Exact historical proof/source

A4b1 Twist into negacyclic ring

WHY: Represent the chirp convolution in the required negacyclic ring.

PORTS: in: SyntheticState; out: SyntheticState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A3b Apply convolution-preserving CRT index map

WHY: Preserve convolution while using the selected multidimensional layout.

PORTS: in: PaddedInputs; out: CRTInputs

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract
SYS-1	Exact historical proof/source		

A4b Chirp source transform

WHY: Realize the source transform through the retained chirp reduction.

PORTS: in: WorkingTensor; out: WorkingSpectrum

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract	
SYS-1	SYS-4	IF-1	IF-5	Exact historical proof/source

A4b2 Two normalized synthetic transforms

WHY: Evaluate the two normalized transforms used by ring convolution.

PORTS: in: SyntheticState; out: SyntheticState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

Full hierarchy | blocks 13-18 of 87

A4b2a Arrange stored tensor layout

WHY: Present stored records in the order required by each transform round.

PORTS: in: SyntheticState; out: SyntheticState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4b2c Apply exact signed polynomial twiddle

WHY: Preserve the exact signed polynomial phase convention.

PORTS: in: SyntheticState; out: SyntheticState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4b3 Packed ring record products

WHY: Form the ring convolution without hiding record-product work.

PORTS: in: SyntheticState; out: SyntheticState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4b2b Execute normalized butterfly rounds

WHY: Realize the normalized local synthetic transform maps.

PORTS: call_B0: BitState; call_C1: ComplexState; in: SyntheticState; out: SyntheticState; receive_B0: BitState; receive_C1: ComplexState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4b2d Truncate round and clear work

WHY: Contain numerical growth and restore temporary work after each round.

PORTS: in: SyntheticState; out: SyntheticState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4b4 Opposite synthetic transform

WHY: Return transformed products through the opposite normalization.

PORTS: in: SyntheticState; out: SyntheticState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

Full hierarchy | blocks 19-24 of 87

A4b4a Opposite rounds and normalization

WHY: Reverse the transform orientation with its exact normalization.

PORTS: in: SyntheticState; out: SyntheticState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4c Gaussian compression

WHY: Return from working-grid values to ordered source samples.

PORTS: in: WorkingSpectrum; out: SourceSpectrum

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-1	SYS-4	IF-1	IF-5	Exact historical proof/source	

A4c2 Apply bounded Neumann correction

WHY: Correct the bounded interpolation discrepancy.

PORTS: in: ResamplingState; out: ResamplingState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4b5 Untwist ring coefficients

WHY: Remove the twist and restore the promised coefficient representation.

PORTS: in: SyntheticState; out: SyntheticState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4c1 Select ordered source samples

WHY: Select the correct ordered source sample set.

PORTS: in: ResamplingState; out: ResamplingState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4c3 Apply diagonal Gaussian correction

WHY: Undo the diagonal Gaussian weighting with a paid factor.

PORTS: in: ResamplingState; out: ResamplingState

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

Full hierarchy | blocks 25-30 of 87

A4d Return ordered source transforms

WHY: Put both operand spectra in the same declared order.

PORTS: in: SourceSpectrum; out: PairedSpectrum

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-1	SYS-4	IF-1	IF-5	Exact historical proof/source	

A4p1 Generate chirp and twist phase descriptors

WHY: Generate phase exponents with their exact address/precision context.

PORTS: in: Descriptors; out: PhaseDescriptor

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4p3 Multiply phase factor and truncate

WHY: Charge factor multiplication and truncation, rather than treating phases as free.

PORTS: in: PhaseApplicationInput; out: WorkingTensor

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4io Adapt records and axis layout

WHY: Make record shape and axis-order adapters explicit.

PORTS: in: WorkingTensor; out: WorkingTensor

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A4p2 Evaluate bounded complex exponential

WHY: Approximate phase factors within the allocated error budget.

PORTS: in: PhaseDescriptor; out: Factors

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract		
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

A5 Multiply matching Fourier entries

WHY: Compute transform-domain convolution products at the chosen precision.

PORTS: in: PairedSpectrum; out: SpectrumProduct

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract
SYS-1	Exact historical proof/source		

Full hierarchy | blocks 31-36 of 87

A6 Apply opposite-sign source transform

WHY: Recover a scaled convolution using the opposite transform sign.

PORTS: in: SpectrumProduct; out: ScaledConvolution

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract
SYS-1	Exact historical proof/source		

A7a First recovery normalization

WHY: Undo the first transform normalization before exact recovery.

PORTS: in: ScaledConvolution; out: PartialRecovery

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract	
SYS-1	SYS-4	IF-5	ARC-4	Exact historical proof/source

A7b Complete recovery normalization and coefficient layout

WHY: Restore every remaining scale and the original coefficient layout.

PORTS: in: PartialRecovery; out: NearIntegerCoefficients

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract	
SYS-1	SYS-4	IF-5	ARC-4	Exact historical proof/source

A8 Round exact integer coefficients

WHY: Use the strict error bound to recover exact integer coefficients.

PORTS: in: NearIntegerCoefficients; out: IntegerCoefficients

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract	
SYS-1	SYS-4	IF-5	ARC-4	Exact historical proof/source

A9 Carry and serialize exact product

WHY: Propagate carries and serialize the exact integer product.

PORTS: in: IntegerCoefficients; out: Product

Retained outer algorithm interface; this package does not execute the all-size transform/recovery internals.

F16	F19	F20	inherited outer contract	
SYS-1	SYS-4	ARC-4	Exact historical proof/source	

B0 Bit address-interchange contract

WHY: Keep F2 payload semantics separate from odd-prime address geometry.

PORTS: in: BitState; out: BitState; return_A4b2b: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F10	F11	F14	F20	current supplier reallocation
SYS-2	IF-1	Exact historical proof/source		

Full hierarchy | blocks 37-42 of 87

B1 Compensate old dirty response

WHY: Cancel arbitrary pre-existing dirty response before desired updates.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F02	F10	current supplier reallocation
IF-2	Exact historical proof/source	

B11 Complete independent dirty frames

WHY: Close arbitrary dirty residuals through the selected exact bank proof.

PORTS: in: BitState; out: BitState

Independent completions are discharged only by the current full-bank proof and exact tags. The historical exterior-completion realization is replaced.

F12	F13	F14	current supplier reallocation	
SYS-2	IF-3	IF-6	ARC-2	Exact historical proof/source

B2 Compute framed helper word

WHY: Produce the local helper action in exact admissible common frames.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F02	F03	F07	F08	current supplier reallocation	
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

B10 Compose five phase-major bit windows

WHY: Compose five routed orientations with legal phase-major boundaries.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F11	current supplier reallocation	
IF-6	ARC-1	Exact historical proof/source

B12 Exchange terminal banks

WHY: Restore the final external data-bank ordering by a paid exchange.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F11	current supplier reallocation
SYS-2	Exact historical proof/source

B3 Manage copied centers

WHY: Use a copied center without mutating its source or reading expired work.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F03	F11	current supplier reallocation
IF-3	Exact historical proof/source	

Full hierarchy | blocks 43-48 of 87

B4 Deliver target updates

WHY: Deliver only the specified target shear while preserving old target values.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F03	F08	F10	current supplier reallocation
IF-2	Exact historical proof/source		

B6a Start source loan

WHY: Enter a source loan with a declared ownership and observation boundary.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F02	current supplier reallocation		
SYS-2	IF-3	Exact historical proof/source	

B6c Use loan alias

WHY: Consume the alias only inside its admitted lifetime and frames.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F02	current supplier reallocation		
SYS-2	IF-3	Exact historical proof/source	

B5 Restore scalar source and dirt

WHY: Return all promised source and dirty payload states.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F10	current supplier reallocation		
SYS-2	IF-2	Exact historical proof/source	

B6b Mix source carrier

WHY: Prepare the source-carrier mix needed by the selected alias recipe.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F02	current supplier reallocation		
SYS-2	IF-3	Exact historical proof/source	

B6d Restore borrowed source

WHY: Return the borrowed source before it is observed by its caller.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F02	F10	current supplier reallocation	
SYS-2	IF-3	Exact historical proof/source	

Full hierarchy | blocks 49-54 of 87

B6e Undo carrier mix

WHY: Remove the carrier mix after its compensation obligations close.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F02	F10	current supplier reallocation
SYS-2	IF-3	Exact historical proof/source

B7b Deliver compensated target response

WHY: Transport the compensated target contribution to its legal observation cut.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F06	F08	current supplier reallocation
IF-2		Exact historical proof/source

B8a Describe full address identity

WHY: Distinguish full routed addresses, not merely equal family numbers.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F11	F14	current supplier reallocation		
SYS-3	IF-1	IF-4	ARC-2	Exact historical proof/source

B7a Prepare source-owned compensation

WHY: Prepare exact compensation for source-owned response.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F02	F06	current supplier reallocation
IF-2		Exact historical proof/source

B7c Keep only desired signal

WHY: Cancel unwanted response without discarding the desired signal.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F06	F08	F10	current supplier reallocation
IF-2			Exact historical proof/source

B8b Execute charged complete-stream route

WHY: Move complete streams with the exact route and finite charge.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F20	current supplier reallocation			
SYS-3	IF-1	IF-4	ARC-2	Exact historical proof/source

Full hierarchy | blocks 55-60 of 87

B9a Call exact nested recursive residual

WHY: Invoke the exact smaller residual operator with restored-row obligations.

PORTS: call_R1: ResidualRequest; in: BitState; out: BitState; receive_R1: ResidualResponse

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F11	F17	F20	current supplier reallocation	
SYS-3	IF-1	IF-4	ARC-3	Exact historical proof/source

B9b Apply equal-frame scalar gate

WHY: Apply payload updates only at their common required frame.

PORTS: in: BitState; out: BitState

Logical responsibility mapped to source527/hybrid. Historical source471 operations and counts are not transplanted one-for-one.

F03	F04	current supplier reallocation	
SYS-3	IF-1	IF-4	Exact historical proof/source

C1 Compensate old complex response

WHY: Cancel the old complex dirty response before the desired linear update.

PORTS: in: ComplexState; out: ComplexState; return_A4b2b: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe	
IF-1	Exact historical proof/source	

C10 Backward transposed-center macro

WHY: Implement the backward/transposed macro with fresh work.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe	
IF-3	Exact historical proof/source	

C11a Copy scalar work

WHY: Create scalar work while preserving its source.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe			
SYS-3	SYS-4	IF-3	IF-5	Exact historical proof/source

C11b Scale exact rational coefficient

WHY: Realize each exact rational coefficient on the declared common grid.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe			
SYS-3	SYS-4	IF-3	IF-5	Exact historical proof/source

Full hierarchy | blocks 61-66 of 87

C11c Accumulate scalar update

WHY: Accumulate the signed scalar response at its correct destination.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe				
SYS-3	SYS-4	IF-3	IF-5	Exact historical proof/source	

C11d Clear scalar work

WHY: Clear temporary scalar work before reuse or recursion.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe				
SYS-3	SYS-4	IF-3	IF-5	Exact historical proof/source	

C12a Route complex owner labels

WHY: Route each complex owner to its exact label and cover class.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe				
ARC-1	Exact historical proof/source				

C12b Compose five complex windows

WHY: Compose all five complex windows, including idle banks.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe				
ARC-1	Exact historical proof/source				

C12c Complete terminal complex exchange

WHY: Finish the signed terminal exchange and restore external ownership.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe				
ARC-1	Exact historical proof/source				

C13 Synthesize phase-correct residual

WHY: Supply the residual with exact complex phases and normalization.

PORTS: call_R1: ResidualRequest; in: ComplexState; out: ComplexState; receive_R1: ResidualResponse

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe				
SYS-3	IF-1	IF-4	ARC-3	Exact historical proof/source	

Full hierarchy | blocks 67-72 of 87

C2 Adapt exact complex entry labels

WHY: Meet the exact phase-sensitive source entry labels.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe
IF-4	Exact historical proof/source

C4 Run retained copied centers

WHY: Retain every copied-center child and scalar consumer.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe				
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

C6 Finish live climbs

WHY: Finish all live climbs before exit adaptation and cleanup.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe
IF-2	Exact historical proof/source

C3 Perform phase A live climbs

WHY: Execute the first set of paid live-frame transitions.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe				
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

C5 Perform phase B live climbs

WHY: Execute the second set of paid live-frame transitions.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe				
SYS-3	IF-1	IF-2	IF-5	IF-6	Exact historical proof/source

C7 Adapt exact complex exit labels

WHY: Meet the exact complex exit labels, not a same-rank substitute.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16	retained complex recipe	
IF-2	IF-4	Exact historical proof/source

Full hierarchy | blocks 73-78 of 87

C8 Restore complex borrowed payload

WHY: Restore borrowed complex payload and clear the tableau.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16retained complex recipe

SYS-2IF-2Exact historical proof/source

C9 Forward copied-center macro

WHY: Implement the literal forward center-copy scalar macro.

PORTS: in: ComplexState; out: ComplexState

Current pinned complex supplier expands this responsibility; use its exact labels, phases, lifetimes and source theorems.

F16retained complex recipe

IF-3Exact historical proof/source

D0 Reconstruct source recipe

WHY: Bind every later check to a reconstructed source recipe.

PORTS: in: Evidence; out: Evidence

Current hybrid proof/verification allocation; historical PASS receipts do not supply current scientific admission.

F02current evidence allocation

SYS-5Exact historical proof/source

D1 Check source pins and preflight

WHY: Reject missing, changed or unpinned inputs.

PORTS: in: Evidence; out: Evidence

Current hybrid proof/verification allocation; historical PASS receipts do not supply current scientific admission.

F01F21current evidence allocation

SYS-5Exact historical proof/source

D2 Check complete bit columns and inverse

WHY: Prove every formal bit column and its inverse restoration.

PORTS: in: Evidence; out: Evidence

Current hybrid proof/verification allocation; historical PASS receipts do not supply current scientific admission.

F10current evidence allocation

SYS-5Exact historical proof/source

D3 Check physical frames routes and primes

WHY: Check actual geometry, route order and determinant inventory.

PORTS: in: Evidence; out: Evidence

Current hybrid proof/verification allocation; historical PASS receipts do not supply current scientific admission.

F11F13F14F15current evidence allocation

SYS-5Exact historical proof/source

Full hierarchy | blocks 79-84 of 87

D4 Check complex splice and declarations

WHY: Bind complex numerical checks to the pinned symbolic declarations.

PORTS: in: Evidence; out: Evidence

Current hybrid proof/verification allocation; historical PASS receipts do not supply current scientific admission.

F16	current evidence allocation
SYS-5	Exact historical proof/source

D6 Expose inherited theorem obligations

WHY: Keep inherited all-size assumptions visible at the certificate boundary.

PORTS: in: Evidence; out: Evidence

Current hybrid proof/verification allocation; historical PASS receipts do not supply current scientific admission.

F21	current evidence allocation
SYS-5	Exact historical proof/source

R2 Call smaller completed transform

WHY: Call a genuinely smaller completed transform under the induction hypothesis.

PORTS: in: ResidualRequest; out: ResidualResponse

Completed-child, fallback and restored-row obligations are retained; current finite bill and exact moments bind their charges.

F17	F18	F20	inherited recursive contract
SYS-3	SYS-5	IF-6	Exact historical proof/source

D5 Check finite bill and exact outer inequalities

WHY: Prevent a local cost improvement from bypassing complete closure.

PORTS: in: Evidence; out: Evidence

Current hybrid proof/verification allocation; historical PASS receipts do not supply current scientific admission.

F17	F18	F19	F20	current evidence allocation
SYS-5	IF-6	ARC-5	Exact historical proof/source	

R1 Form completed residual request

WHY: Package the complete residual operator and ownership obligations.

PORTS: in: ResidualRequest; out: ResidualResponse; return_B9a: ResidualResponse; return_C13: ResidualResponse

Completed-child, fallback and restored-row obligations are retained; current finite bill and exact moments bind their charges.

F17	F18	F20	inherited recursive contract
SYS-3	SYS-5	IF-6	Exact historical proof/source

R3 Stop at attained ordinary leaf

WHY: Terminate at an attained ordinary algorithm below the finite cutoff.

PORTS: in: ResidualRequest; out: ResidualResponse

Completed-child, fallback and restored-row obligations are retained; current finite bill and exact moments bind their charges.

F17	F18	F20	inherited recursive contract	
SYS-3	SYS-5	IF-6	ARC-5	Exact historical proof/source

Full hierarchy | blocks 85-87 of 87

R4 Restore borrowed high rows

WHY: Return every borrowed high row and spectator state.

PORTS: in: ResidualRequest; out: ResidualResponse

Completed-child, fallback and restored-row obligations are retained; current finite bill and exact moments bind their charges.

F17	F18	F20	inherited recursive contract			
SYS-2	SYS-3	SYS-5	IF-3	IF-6	ARC-5	Exact historical proof/source

R5a Dispatch good and bad q-local edges

WHY: Retain both good-edge dispatch and the complete bad-edge fallback.

PORTS: in: ResidualRequest; out: ResidualResponse

Completed-child, fallback and restored-row obligations are retained; current finite bill and exact moments bind their charges.

F17	F18	F20	inherited recursive contract			
SYS-3	SYS-5	IF-4	IF-6	ARC-3	Exact historical proof/source	

R5b Repair exceptional layout records

WHY: Repair exceptional record layouts with paid primitives.

PORTS: in: ResidualRequest; out: ResidualResponse

Completed-child, fallback and restored-row obligations are retained; current finite bill and exact moments bind their charges.

F17	F18	F20	inherited recursive contract			
SYS-3	SYS-5	IF-4	IF-6	Exact historical proof/source		

Preserved interface connections | 1-10 of 53

HT001	A0a.out retained outer contract	->	A0b.in	data / Descriptors
HT002	A0b.out retained outer contract	->	A0c.in	data / Descriptors
HT003	A0c.out retained outer contract	->	A0d.in	data / Descriptors
HT004	A0d.out retained outer contract	->	A1.in	data / Descriptors
HT005	A1.out retained outer contract	->	A2.in	data / Digits
HT006	A2.out retained outer contract	->	A3a.in	data / ScaledInputs
HT007	A3a.out retained outer contract	->	A3b.in	data / PaddedInputs
HT008	A3b.out retained outer contract	->	A4a.in	data / CRTInputs
HT009	A4a.out retained outer contract	->	A4b.in	data / WorkingTensor
HT010	A4b.out retained outer contract	->	A4c.in	data / WorkingSpectrum

Also preserved: 75 documentary relations with separate classification in hierarchy_edges.json. Neither inventory is silently relabeled as current gate order.

Current F01-F21 dependencies and the exact selected local event inventory are separate, source-bound layers.

Preserved interface connections | 11-20 of 53

HT011	A4c.out retained outer contract	->	A4d.in	data / SourceSpectrum
HT012	A4d.out retained outer contract	->	A5.in	data / PairedSpectrum
HT013	A5.out retained outer contract	->	A6.in	data / SpectrumProduct
HT014	A6.out retained outer contract	->	A7a.in	data / ScaledConvolution
HT015	A7a.out retained outer contract	->	A7b.in	data / PartialRecovery
HT016	A7b.out retained outer contract	->	A8.in	data / NearIntegerCoefficients
HT017	A8.out retained outer contract	->	A9.in	data / IntegerCoefficients
HT018	A4b1.out retained outer contract	->	A4b2.in	data / SyntheticState
HT019	A4b2.out retained outer contract	->	A4b3.in	data / SyntheticState
HT020	A4b3.out retained outer contract	->	A4b4.in	data / SyntheticState

Also preserved: 75 documentary relations with separate classification in hierarchy_edges.json. Neither inventory is silently relabeled as current gate order.

Current F01-F21 dependencies and the exact selected local event inventory are separate, source-bound layers.

Preserved interface connections | 21-30 of 53

HT021	A4b4.out retained outer contract	->	A4b5.in	data / SyntheticState
HT022	A4b2a.out retained outer contract	->	A4b2b.in	data / SyntheticState
HT023	A4b2b.out retained outer contract	->	A4b2c.in	data / SyntheticState
HT024	A4b2c.out retained outer contract	->	A4b2d.in	data / SyntheticState
HT025	A4c1.out retained outer contract	->	A4c2.in	data / ResamplingState
HT026	A4c2.out retained outer contract	->	A4c3.in	data / ResamplingState
HT027	B1.out historical order replaced	->	B2.in	data / BitState
HT028	B2.out historical order replaced	->	B3.in	data / BitState
HT029	B3.out historical order replaced	->	B4.in	data / BitState
HT030	B4.out historical order replaced	->	B5.in	data / BitState

Also preserved: 75 documentary relations with separate classification in hierarchy_edges.json. Neither inventory is silently relabeled as current gate order.

Current F01-F21 dependencies and the exact selected local event inventory are separate, source-bound layers.

Preserved interface connections | 31-40 of 53

HT031	B8a.out historical order replaced	->	B8b.in	data / BitState
HT032	B8b.out historical order replaced	->	B9a.in	data / BitState
HT033	B9a.out historical order replaced	->	B9b.in	data / BitState
HT034	B9b.out historical order replaced	->	B10.in	data / BitState
HT035	B10.out historical order replaced	->	B11.in	data / BitState
HT036	B11.out historical order replaced	->	B12.in	data / BitState
HT037	C1.out retained complex interface	->	C2.in	data / ComplexState
HT038	C2.out retained complex interface	->	C3.in	data / ComplexState
HT039	C3.out retained complex interface	->	C4.in	data / ComplexState
HT040	C4.out retained complex interface	->	C5.in	data / ComplexState

Also preserved: 75 documentary relations with separate classification in hierarchy_edges.json. Neither inventory is silently relabeled as current gate order.

Current F01-F21 dependencies and the exact selected local event inventory are separate, source-bound layers.

Preserved interface connections | 41-50 of 53

HT041	C5.out retained complex interface	->	C6.in	data / ComplexState
HT042	C6.out retained complex interface	->	C7.in	data / ComplexState
HT043	C7.out retained complex interface	->	C8.in	data / ComplexState
HT044	C12a.out retained complex interface	->	C12b.in	data / ComplexState
HT045	C12b.out retained complex interface	->	C12c.in	data / ComplexState
HT046	A4b2b.call_B0 retained bit interface contract	->	B0.in	call / BitState
HT047	B0.return_A4b2b retained bit interface contract	->	A4b2b.receive_B0	return / BitState
HT048	A4b2b.call_C1 retained complex interface	->	C1.in	call / ComplexState
HT049	C1.return_A4b2b retained complex interface	->	A4b2b.receive_C1	return / ComplexState
HT050	B9a.call_R1 retained bit interface contract	->	R1.in	call / ResidualRequest

Also preserved: 75 documentary relations with separate classification in hierarchy_edges.json. Neither inventory is silently relabeled as current gate order.

Current F01-F21 dependencies and the exact selected local event inventory are separate, source-bound layers.

Preserved interface connections | 51-53 of 53

HT051	R1.return_B9a retained bit interface contract	->	B9a.receive_R1	return / ResidualResponse
HT052	C13.call_R1 inherited recursive contract	->	R1.in	call / ResidualRequest
HT053	R1.return_C13 inherited recursive contract	->	C13.receive_R1	return / ResidualResponse

Also preserved: 75 documentary relations with separate classification in hierarchy_edges.json. Neither inventory is silently relabeled as current gate order.

Current F01-F21 dependencies and the exact selected local event inventory are separate, source-bound layers.

Original requirement vocabulary | 1-5

SYS-1 Exact integer product

For every allowed input, output represents the exact integer product.

Why: This is the requested mathematical result.

SYS-2 Honor promised state and observations

Preserve caller-promised values and properties at every required observation.

Why: Endpoint-only equality can hide corrupted intermediate users.

SYS-3 Realizable execution in the declared model

Implement every operation and movement claimed in the stated model with legal resources.

Why: Algebra alone is not a physical cost proof.

SYS-4 Justified exact recovery

Prove every numerical approximation affecting the result is compatible with exact decoding.

Why: Local exact algebra does not bound accumulated numerical error.

SYS-5 Complete support for the claimed cost

Account for all resources relevant to the claimed result and bind evidence to exact sources and scope.

Why: Omitted costs and conflated evidence can reverse a comparison.

Original requirement vocabulary | 6-10

IF-1 Declared mathematical domains

Specify payload/address domains, sizes, legal inputs and inherited hypotheses.

Why: Equal notation across fields may denote different maps.

IF-2 Complete boundary and observations

Specify endpoint and every required observation at semantic events.

Why: An unobserved intermediate coordinate need not be fixed.

IF-3 Ownership and lifetimes

Declare owners, live versions, dirty/old state, loans, retirement and copy/erase conditions in both directions.

Why: Forward last-use is not inverse restoration.

IF-4 Physical representations and routes

Specify exact frames/labels and complete address action, spectators, inverse and simultaneous injectivity.

Why: Equal ranks or small address-matrix similarity are insufficient.

IF-5 Consumed numerical envelope

Bound coefficients, signed growth, norms, errors and record widths consumed by callers.

Why: A new representative may be valid but have different growth.

Original requirement vocabulary | 11-15

IF-6 Complete paid resource interface

Declare net child inventory, true stock, replicas, finite costs, fallback, shrink and normalization.

Why: Lower call count is not equivalent to lower paid moment.

ARC-1 Retained phase-major composition

When retaining the current global theorem, respect complete-cover phase barriers and dependencies.

Why: Interleaving may expose unfinished state.

ARC-2 Independent completed bank realization

When using independent banks, actual routed block projectors are disjoint and sum to each required bank endpoint.

Why: Volume and ranks alone do not complete a bank.

ARC-3 Eligible chart arithmetic

For the rational-chart reduction, every consumed inverse/scaling exists at each eligible prime/ring.

Why: Invalid denominators make routing undefined.

ARC-4 Retained approximate convolution recovery

For this recovery, restore both normalizations and coefficient order and achieve pre-rounding error below one half.

Why: The decoder's input assumptions must be established.

Original requirement vocabulary | 16-16

ARC-5 Selected ordinary-leaf and outer assembly

For this bound, use attained ordinary savings and valid bridge data in the selected outer inequalities.

Why: A limiting coarse rate is not a completed leaf.

Requirement traceability | 1-6

R01 Immutable construction

All numerical code, witness bytes and exact source pins must remain bound to one manifest.

F01 F02 F07 F09 F21

R02 Payload and restoration

All arbitrary input, old-target and dirty columns satisfy the F2 supplier contract and reverse restoration.

F02 F03 F04 F06 F07 F08 F10 F11 F14

R03 Legal actual geometry

Every used frame, inclusion, complement, chart and boundary must be checked as an exact subspace/operator, not by rank alone.

F02 F03 F04 F05 F06 F07 F08 F09 F11 F13 F15

R04 Complete physical price

Every recursive child, scalar primitive, move, copy, bridge, route, selector and fallback must remain accounted for.

F03 F04 F05 F06 F07 F08 F09 F11 F12 F13 F14 F16 F17 F20

R05 Full bank address safety

Every literal helper occurrence is assigned once; complete banks cover all coordinates; full addresses are nonaliasing.

F11 F12 F13 F14

R06 Separate lift and precision claims

F2 semantic equality and chosen signed-lift prefix majorants remain distinct from the unchanged source's signed decoder.

F04 F08 F10 F16 F20

Requirement traceability | 7-12

R07 Complete prime coverage

Every physically consumed and defining-source basis has exact determinant/exclusion evidence for eligible odd q.

F13 F15 F20

R08 Strict directed moment gap

Both interval engines and the full rare fallback establish strict moment bounds and reject the adjacent grid.

F17 F18 F20

R09 Finite ordinary closure

Use three finite bootstrap levels, all outer inequalities and the full inherited-constant cutoff, with old-leaf fallback.

F17 F18 F19 F20

R10 Inherited theorem boundary

All-size compiler, covers/charts, restored rows, routing, prime supply, recovery, complex-symbolic and analytic hypotheses stay explicit.

F16 F19

R11 Evidence honesty

Describe which stages ran freshly, what unchanged evidence was reused, and which cross-platform checks remain pending.

F01 F10 F15 F16 F21

R12 Documentation provenance

Preserve historical artifacts, licenses and contributor credit; describe the selected construction rather than additive historical gains.

F21

Source anchors | 1-8

S01 verify.py : 26

```
def integrity()  
SHA256 489bee6d23fbed6732e60d62f8fa98e7d2dfb2d2b7f75fd2bd3bc4a920155f10
```

S02 PR259-PROVENANCE.md : 1

```
# PR259 lineage  
SHA256 a232f3fb631976128d4e454dc414ecae7de1e88fba86ca45d13e4af21d8bb5f8
```

S03 prepare.py : 12

```
def prepare()  
SHA256 fc109fe81d7d9d3f53483ce7d648dc9fcf6d9fb71b8fde1c867ece5283fadbcc
```

S04 source527/word.py : 1

```
def  
SHA256 e675d4eb9d90ff0b44279fae0b46f1a8b39f65421cc38454b70a6b75d0e10b42
```

S05 code/physical527.py : 11

```
def run(  
SHA256 1723a5c53ec236529e46980c8eb2c45134c61275ac7fb85c8ad92223784de07c
```

S06 parity_transform.py : 62

```
def run(  
SHA256 ef6f5224b91c409c30fc0886143787a0839a13cc6b148e42ca5db858d502ec4d
```

S07 proof/PARITY-CONTRACT.md : 20

```
## Operator proof  
SHA256 a9246be656e31f1a21d5ec714e002f51be0aa07c36e1528066a25b2c7a02db91
```

S08 retiming_transform.py : 65

```
def run(  
SHA256 483cb8da83a1444cbc8b1bbe60473e571a27e6e589fa64f9a26910ca0d678407
```

Source anchors | 9-16

S09 raw_ledger.py : 190

def rebind_retiming(
SHA256 d5dae7b56d403adb778251e4b42903695632d522c1392505723e2125162cef78

S10 regauge_transform.py : 128

def run(
SHA256 baa1c7390f449722e7e700fa1ce9afc3d00cd4c0f7faa5f41693ff88ec02fd61

S11 raw_ledger.py : 216

def rebind_regauge(
SHA256 d5dae7b56d403adb778251e4b42903695632d522c1392505723e2125162cef78

S12 hybrid_transform.py : 19

def run(
SHA256 c06837bbc6ccf068362e1f09f110319f20135588f42e3cb3f8bc753436802e09

S13 hybrid-selection.json : 4

"removed_pivots"
SHA256 0229c2fad3b448d33ad1f21900ad67d13ab1e0c8e716ced9ec66b4ab0445c0b1

S14 PR259-PROVENANCE.md : 3

PR249:
SHA256 a232f3fb631976128d4e454dc414ecae7de1e88fba86ca45d13e4af21d8bb5f8

S15 hybrid_transform.py : 92

local=[
SHA256 c06837bbc6ccf068362e1f09f110319f20135588f42e3cb3f8bc753436802e09

S16 PROOF.md : 16

Kernel and observation boundaries
SHA256 d7c10b19fd1e5fe9280e09c9848bbd7c1c53b36363e6b98571ae8513932ad843

Source anchors | 17-24

S17 retiming25.py : 22

def transform(
SHA256 5e9e835730aac1b0752019530daa1e75c8938686bf8a8a01640ab69ceada90f2

S18 hybrid_transform.py : 146

out,retiming=
SHA256 c06837bbc6ccf068362e1f09f110319f20135588f42e3cb3f8bc753436802e09

S19 scalar_check.py : 30

def replay_projection(
SHA256 13ac9a3ef3c131739478ac1ba5903acd638cf5a41c6fa463fa3e27325413c7f9

S20 scalar_check.py : 54

def run(
SHA256 13ac9a3ef3c131739478ac1ba5903acd638cf5a41c6fa463fa3e27325413c7f9

S21 code/global_lowering.py : 66

class Lowerer:
SHA256 c888c4d3c38d4a59b48f8a328fc495b3b38ee36ede133495522d8d76c100794b

S22 portable_bit.py : 15

def phase_major_templates(
SHA256 51760a4ba5b431d55b245aca2036ef51249942503fdcd5d9317d515ce411b87c

S23 bank_template.py : 11

class BankPlan:
SHA256 e255bdc0c0aa4465485ee34ece7d1f04ce3ce363d8ac62edc9de5608e46f8965

S24 bank_check.py : 49

def census(
SHA256 64790d1d45032e7f8ae2252d592759ff82def8033544107f9f0dd3bbc6841581

Source anchors | 25-32

S25 BANK-PROOF.md : 9

Literal capacity
SHA256 ae2ca2693d75f56c076c2ffc790163fe23d2bf69de23bed94345cfe48a0b4252

S26 bank_check.py : 97

def run(
SHA256 64790d1d45032e7f8ae2252d592759ff82def8033544107f9f0dd3bbc6841581

S27 bank_template.py : 68

def normalizer(
SHA256 e255bdc0c0aa4465485ee34ece7d1f04ce3ce363d8ac62edc9de5608e46f8965

S28 BANK-PROOF.md : 39

Generic charts
SHA256 ae2ca2693d75f56c076c2ffc790163fe23d2bf69de23bed94345cfe48a0b4252

S29 bank_template.py : 130

def map_stage_row(
SHA256 e255bdc0c0aa4465485ee34ece7d1f04ce3ce363d8ac62edc9de5608e46f8965

S30 bank_template.py : 146

def map_boundary_row(
SHA256 e255bdc0c0aa4465485ee34ece7d1f04ce3ce363d8ac62edc9de5608e46f8965

S31 bank_check.py : 65

class BankLowerer:
SHA256 64790d1d45032e7f8ae2252d592759ff82def8033544107f9f0dd3bbc6841581

S32 code/geometry527.py : 11

def run(
SHA256 bff96140e93c0cea01ed618b49b913f913c12d99ebb0e4e4342336cf5a102800

Source anchors | 33-40

S33 prime_check.py : 20

```
def run(  
SHA256 80b764f53b92b299569a401754047b1a86f80ba4046502bfbfea924f6ec95e0e
```

S34 portable_complex.py : 77

```
def run(  
SHA256 797e144c0040d78a4731a6eaa61c0edb9803a08462f885ed014cbca9d9991d82
```

S35 portable_complex.py : 25

```
def finite_guard(  
SHA256 797e144c0040d78a4731a6eaa61c0edb9803a08462f885ed014cbca9d9991d82
```

S36 math_check.py : 23

```
def run(  
SHA256 bb7c80571104d9b9c30ce7be53654090dc006583ba1838557a5307b8a9db4a39
```

S37 moment.py : 55

```
def moment(  
SHA256 9d465c716c6b910ff66e1453e3bf0057608a787b5f5361be93781f42cb3370f7
```

S38 base_two_moment.py : 52

```
def moment(  
SHA256 aee70c242a2f1d225528fabe2002e1564d1dc884797c07d28a831c7878ee9098
```

S39 math_check.py : 46

```
chain=[  
SHA256 bb7c80571104d9b9c30ce7be53654090dc006583ba1838557a5307b8a9db4a39
```

S40 finite_check.py : 130

```
chain=[  
SHA256 dfb8dd62c4d068417434f6a43e15e5f69dfa40b4d95b7a8d4995b6950c095c61
```

Source anchors | 41-46

S41 outer.py : 16

```
def assembly(  
SHA256 c1ebb52163a7219504c728a5c2b098bf1e50bf20148a2879c5a74c3cad7c03c
```

S42 math_check.py : 53

```
assembly=outer.assembly(  
SHA256 bb7c80571104d9b9c30ce7be53654090dc006583ba1838557a5307b8a9db4a39
```

S43 finite_check.py : 59

```
def run(  
SHA256 dfb8dd62c4d068417434f6a43e15e5f69dfa40b4d95b7a8d4995b6950c095c61
```

S44 finite_check.py : 17

```
def cutoff_log2(  
SHA256 dfb8dd62c4d068417434f6a43e15e5f69dfa40b4d95b7a8d4995b6950c095c61
```

S45 verify.py : 50

```
def main()  
SHA256 489bee6d23fbed6732e60d62f8fa98e7d2dfb2d2b7f75fd2bd3bc4a920155f10
```

S46 verify.py : 37

```
def validate_required(  
SHA256 489bee6d23fbed6732e60d62f8fa98e7d2dfb2d2b7f75fd2bd3bc4a920155f10
```

Coverage and publication boundary

Selected active heritage

PR249's 21 transported entrances -> PR259's 513 retained cohorts (five quartets removed) + seven new delayed-inverse intervals -> 25 same-gate PR258/PR260 retimings once. PR254 is ancestry, not another additive saving. PR251's extra sixteen entrances are excluded.

Architectural and theorem heritage

PR234 supplies Henry Grant / hcg890's five-stage architecture and finite interfaces. PR237 is a prior width-120 bank extension; no exclusive priority is claimed. The source527 and completed-bank chains retain all upstream licenses and contributor notices. Complex source is Jacob Sussman's f010392 snapshot; PR209 is its documentation pointer, and PR193 supplies the helper lineage.

Coverage stated exactly

The current architecture catalog and dependency index account for every declared block input port. All 87 original hierarchy IDs are explicitly reconciled. The visual views summarize architecture and compiler flow; reference pages preserve source and requirement traceability. Inherited theorem internals remain unexpanded.

Evidence and pending checks

The recorded cloud admission freshly executed changed stages and reused byte/AST-bound unchanged signed-source and complex evidence. Mac standalone replay has now reported all eight stages passing with 143 unchanged source files. This explicit PR259 hybrid checkpoint remains mathematically conditional and is below the later PR266 frontier. Documentation checks are separate from scientific replay.

Public follow-up only

Construction checkpoint PR #277: <https://github.com/CrocSwap/integer-mult-bounds/pull/277>. Public source commit c2f07d311bfdad0d89bf961e4da133ee97bb78de is linked in README.md. Source anchors retain exact relative paths, symbols, line numbers and hashes. The historical combined_system_schematic_20261010 generator is preserved unchanged.

Reproduce the architecture reference

Install

Python 3.11+; `python -m pip install -r joint259_system_docs_requirements.txt`. Pinned reportlab 4.4.9, pypdf 6.10.0, Pillow 12.3.0, charset-normalizer 3.5.1. Graphviz is not used or required.

Architecture build command

`python -B joint259_system_docs_incremental.py --source /path/to/public-package --output joint259-docs-build --svg`. An admitted source ZIP also works. Actual consumed manifest is recorded separately from the tested science archive in `source_binding.json` and `QA.json`.

Outputs

`functional_architecture.pdf`; `summary-functional.svg`; `QA.json`; `incremental_receipt.json`. The retained renderer also emits local source, requirement and dependency indexes from the same pinned source.

Automatic reproducibility checks

Build every PDF twice in memory and require byte-identical SHA256, unchanged manifest, no missing/extra source files, valid proof anchors, all catalog block IDs present, all internal links resolved, nonempty extractable page text and no layout overflow. `QA.json` records counts, pins and output digests.

Independent visual check

Use Poppler: `pdftoppm -png -r 100 functional_architecture.pdf review/page`. Inspect every page for clipping, missing glyphs and readable connectors. Poppler is an optional renderer, not a generation dependency; record its actual version in visual QA.

Scientific verification remains separate

From a clean source package, use Python 3.11+ and SymPy 1.14.0 with assertions enabled: `python -B verify.py --output /new/output/directory`. Never write inside the immutable source package; never use `-O`. A successful documentation build is not an all-eight-stage science replay.